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Scope of Work

Siemens proposes the following:

Furnish and install the Siemens Building Management System (BMS) in accordance with the drawings and specifications by AMA Consulting Engineers.

- Mechanical plans dated 4/03/2026 - Issued for pricing
- Specification on plans
- Mechanical plans dated 4/22/2026 - Addendum #1

Detailed Scope of Work

The following list of equipment will be controlled and/or monitored by the BAS:

Qty	System Description
3	PXC BMS Primary Controllers
93	Fan Powered Boxes
3	Water Cooled AC units
6	CW Pumps
5	Fire Smoke Dampers
11	Pantry Sink / closet Leak Detectors
6	HW Heater Leak Detectors
3	Tenant Auto Dialers
	Alternate pricing - See below

The following equipment/systems will be controlled and/or monitored by the Siemens BMS as described. For counts per area refer to the above list of equipment per the breakdown:

General (building standards)

-
  - Use of plenum rated cabling for low voltage in open ceilings
  - Orange FLN / Comm Wire
  - Thermostat – Sensing only
  - 2/3 Static pressure resides in AC unit - run out Double gang poly and tip
  - Only tenant leak alarms to BMS. All other tenant equipment alarms to tenant Autodialer.
  - Leak Detector Liebert LT-410 only (building standard)
  - Laptop for ABT project on site. Use Laptop for all DXR projects. Contact Siemens CSM.
  - Field coordinate with Fire alarm contractor to provide new Fan Powered Box Shut Down contact at Mammoth AC unit DDC panel.

- **OPEN SHOP Electrical Subcontractor**

PXC DDC Controller

- Siemens to replace / upgrade existing primary DDC controller to enable communication to new FPB controllers via BACnet MSTP.
  - Re- connect existing BLN to new controller
  - Install new Ethernet cable to adjacent Base Building IT closet and safe-off for future connection (BB IT contractor to install future network switch).
  - Fire Alarm contractor to provide new shut down / status relays within 3ft of controllers
  - Re-install new differential pressure transmitter for existing AHU (removed during demo).

Water Cooled AC Units

- Water cooled AC units to be provided with manufacturer controls. Siemens to provide wiring and devices as follows:
  - Interlock wiring between CRAC and WCCU
  - Install and wire manufacturer's temperature/humidity room sensor
  - Hardwire interlock unit operation to pumps
  - Furnish (2) Isolation valves
  - Furnish and wire (2) Leak detectors (1-Condesing Unit / 1-to alarm at Base Building BMS and Tenant Autodialer. Leak detectors to close isolation valves upon alarm.

FP Boxes w/ Hot Water Reheat

- Siemens to provide factory mounted controls on FP boxes, and field FLN communication wiring. The following is a list of the factory mounted controls, monitored and controlled by Siemens:
  - (1) DXR BACnet MSTP equipment controller with damper actuator
  - (1) Fan - Start/stop and status
    - Current sensor p/n: ECMCS (Automation Components)
  - (1) Room Temperature Sensor – Sensing Only
  - (1) Discharge Air Temperature Sensor
  - (1) Hot Water Valve and Actuator
  - **Add Freeze protection program per Building standard.**
  - **Add Unoccupied mode program: Dampers full-open to prevent Mammoth unit shut down.**

CW Pumps

- Siemens to wire pumps and associated sensors to local lead / lag panel (provided by others).
  - (1) Install and power manufacturer provided Lead / Lag panel
  - (2) Isolation valves - furnish and wire
  - (2) Leak detectors - furnish and wire. To alarm at BB BMS and Tenant AutoDialer
  - (2) Flow Switches - furnish and wire
  - (2) Pump - Start/stop and Status
  - (1) Differential pressure transmitter - to maintain pump speed
  - (1) Hardwire interlock to AC unit operation

- **No connection to base building BMS**
- **Lead Lag panel provided by others**

Fire Smoke Dampers

- Siemens to furnish pneumatic labor to connect pneumatic tubing to new FSDs and existing Fire Alarm contacts. FSD dampers to operate in conjunction with existing FSDs serving Mammoth Unit

Leak Detectors - Pantry Sink and HW Heaters

- Siemens to furnish and install Leak detectors. Hardwire interlock to isolation valve (valve furnished, installed, and powered by others).
  - Leak detectors to alarm at BB BMS and Tenant AutoDialer

Autodialer

- Siemens to furnish and install tenant autodialer. Tenant to provide IP connection.

Alternate Price

1. ~~**Add $125 per Fan Powered Box -** (1) Field install controller in lieu of factory mounting.~~
2. ~~**Add $575 per CO2 room Sensor -** if required. Specification calls for CO2 monitoring in large areas, but plans do not show CO2 room sensors…. only show room temperature sensors.~~
3. ~~**Add $2,800 per Panel -** Furnish (1) pump Lead / Lag DDC panel with software. Stand-alone. No tie-in to BMS~~
4. ~~**Add $62,200 -** Conduit stub-ups and backboxes for thermostat installation (if required)~~
5. **Deduct $84,300** - Open shop labor for electrical labor **ACCEPTED and INCLUDED**

Exclusions

1. Conduit stub-ups and backboxes for thermostats
2. Conduit in open ceilings for low voltage & communication wiring (cable/wire to be plenum rated)
3. Instantaneous hot water heaters (i.e. serving eye wash stations)
4. PRV Stations, water meters, electric meters.
5. Furnishing Lead / lag panels for pumps
6. Base building BMS monitoring/control of (except Leak Detection):
    1. Pumps
    2. Water Cooled Computer room AC units
    3. Transfer Fan
    4. **Exhaust fan Dish Room**
7. Field installation of Fan Powered Box controls
8. Furnishing plumbing Isolation valves (i.e. HW heaters, sinks, etc..)
9. Air Purification System
10. CO2 sensors (none shown on plans)
11. Overtime
12. Per project aggregate general liability insurance
13. 120VAC Power wiring to all terminal box controllers
14. Furnishing, installing, or wiring of duct smoke detectors
15. Wiring for fire alarm shutdown of Fan Powered Box and/or CW AC equipment
16. Furnishing CW AC thermostats
17. FPB box control transformers
18. Installation of pipe mounted devices (i.e. control valves, flow meters, sensor wells)
19. Access doors
20. Temporary or standby labor and services

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Clarifications

1. Cost for Phasing and/or Escalation has not been included. Project completion estimated within twelve (12) calendar months from reception of Purchase order / contract.
2. Price based on a **Capital Improvement** project. Appropriate taxes will be added to final sale price if certificate is not provided.
3. Sample Certificate of Insurance must be provided with contract / purchase order. Required Insurance must be listed in commitment document.
4. **Estimated durations** are as follows from acceptance of PO / Contract:
    1. Purchase order / contract processing to activate Job number (internal) - 6 to 8 business days
    2. Submittal - 1st submission - 2 to 4 weeks from job activation
    3. On site installation to begin - upon approval of submittals
    4. Material shipments are experiencing unusually long lead times
5. The project cost includes the tie in of the associated BMS system to the existing Siemens BMS front end workstation
6. Graphic floorplan can only be generated from the Autocad files provided. Please provide the correct floorplan area(s) (full/partial) required to be displayed on the BMS.
7. Control wiring shall be run in conduit where exposed to outdoor conditions. Control wiring shall be run using plenum rated cable without conduit in all other locations
8. Miscellaneous wiring or power to 3rd party equipment has not been included, unless noted in Scope of Work section of this proposal.
9. If Per project Aggregate insurance is required: Add 1.1% to final sale (Per Aggregate Insurance would be noted in Cert. Of Insurance provided by GC / Building owner).